CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

IO Organization

Lecture No.- 2



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Recap of Previous Lecture



Topic Peripheral Device

Topic Asynchronous Serial Data Transfer

Topics to be Covered



- **Topic** Programmed IO
- **Topic** Interrupt IO
- **Topic** DMA

[MCQ] [2025 : 1M]

#Q. Suppose a program is running on a non-pipelined single processor computer system. The computer is connected to an external device that can interrupt the processor asynchronously. The processor needs to execute the interrupt service routine (ISR) to serve this interrupt. The following steps (not necessarily in order) are taken by the processor when the interrupt arrives:

(i) The processor saves the content of the program counter.

(ii) The program counter is loaded with the start address of the ISR.

(iii) The processor finishes the present instruction.

Which ONE of the following is the CORRECT sequence of steps?

A (iii), (i), (ii)

B (i), (iii), (ii)

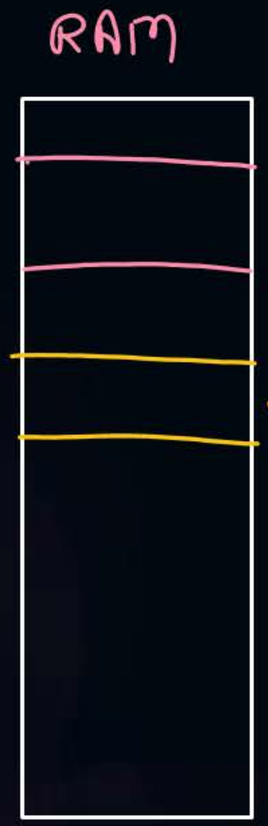
C (i), (ii), (iii)

D (iii), (ii), (i)



Topic : Interrupt Initiated IO





← execution of which services the interrupt



Topic : Vectored vs Non-Vectored

(scalar interrupt)

Device sends interrupt and

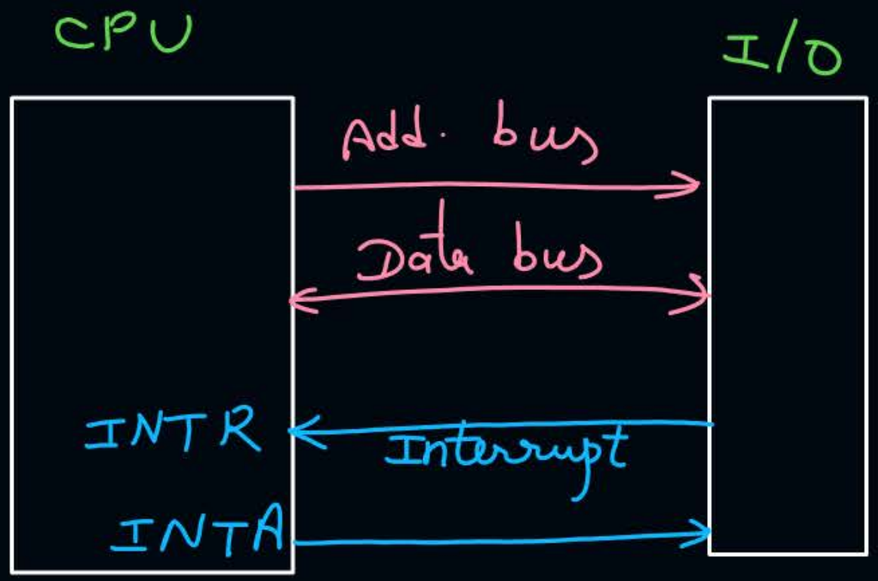
Vector both.

→ address or reference of ISR

Device sends only interrupt

↓

CPU runs a default service routine, and figures out add. of actual ISR.



INTR ⇒ Interrupt Request

INTA ⇒ —11— Acknowledgement

Sequence :-

I/O to CPU ⇒ INTR } Control signals

CPU to I/O ⇒ INTA }

I/O to CPU ⇒ vector } through data bus



Topic : Maskable vs Non-Maskable

CPU can accept or reject

CPU accepts always



Topic : Internal Vs External

if CPU is interrupted
due to unexpected error
during an instruction execution.

ex:- page fault

if device generates interrupt



Topic : Simultaneous Interrupts

If multiple devices generate interrupt simultaneously then highest priority device's interrupt will be serviced first.

Priority based interrupt handling

- S/w solution
- H/w solution
  - Serial (daisy chaining)
  - Parallel

Hard disk gets highest priority



Topic : Time Required in Interrupt IO

= Interrupt overhead time + Interrupt service time

[NAT]

#Q. Consider a CPU which takes 0.05 microseconds as interrupt overhead time when a device generates interrupt for CPU, and CPU accepts it. After that CPU takes 6 cycles to service the interrupt. If CPU runs on 10MHz clock rate then total time CPU spends for interrupt service is _____ microseconds?

$= 0.05\ \mu sec + 6\ cycles$

$= 0.05\ \mu sec + 6 * \frac{1}{10\ MHz}$

$= 0.05\ \mu sec + 6 * 0.1\ \mu sec$

$= 0.65\ \mu sec$

[NAT]

#Q. A device with data transfer rate 20 KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 10 microsecond.

1. Total time required in programmed IO for 10 bytes data transfer? 550 μsec
2. Total time required in interrupt IO for 10 bytes data transfer? 510 μsec
3. What is the minimum performance gain of operating the device under interrupt mode over operating it under program controlled mode?

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1. = status check time + data transfer time

= 1B status transfer time from I/O to CPU + 10B data transfer time

= 50 μsec + 10 * 50 μsec = 550 μsec

for 20KB, device takes = 1sec

for 1B, ——11—— = $\frac{1\text{sec}}{20\text{KB}} * 1\text{B}$

$= 0.05\text{msec}$

$= 50\ \mu\text{sec}$

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2. $= 10\ \mu\text{sec} + 10 * 50\ \mu\text{sec}$

$= 510\ \mu\text{sec}$

3. speed up or performance gain $= \dfrac{\text{slower technique time}}{\text{faster technique time}}$

interrupt I/O performance $= \dfrac{1}{510 \text{ usec}}$

Prog. I/O performance $= \dfrac{1}{550 \text{ usec}}$

$= \dfrac{\frac{1}{510}}{\frac{1}{550}} = \dfrac{550}{510}$

$= \dfrac{550 \text{ usec}}{510 \text{ usec}}$

$= 1.078$

$= 1.08$

[NAT]

Ans = 25

GATE – PYQ

#Q. A device with data transfer rate 10 KB/sec is connected to a CPU. Data is transferred byte-wise. Let the interrupt overhead be 4 microsecond. The byte transfer time between the device interface register and CPU or memory is negligible. What is the minimum performance gain of operating the device under interrupt mode over operating it under program controlled mode?

$$\text{Performance gain} = \frac{100 + 0}{4 + 0}$$

$$= 25$$

for 10 KB, time = 1 sec

for 1 B, time = $\frac{1\text{ sec}}{10\text{ KB}} * 1\text{ B}$

$= 0.1$ msec

$= 100$ μsec

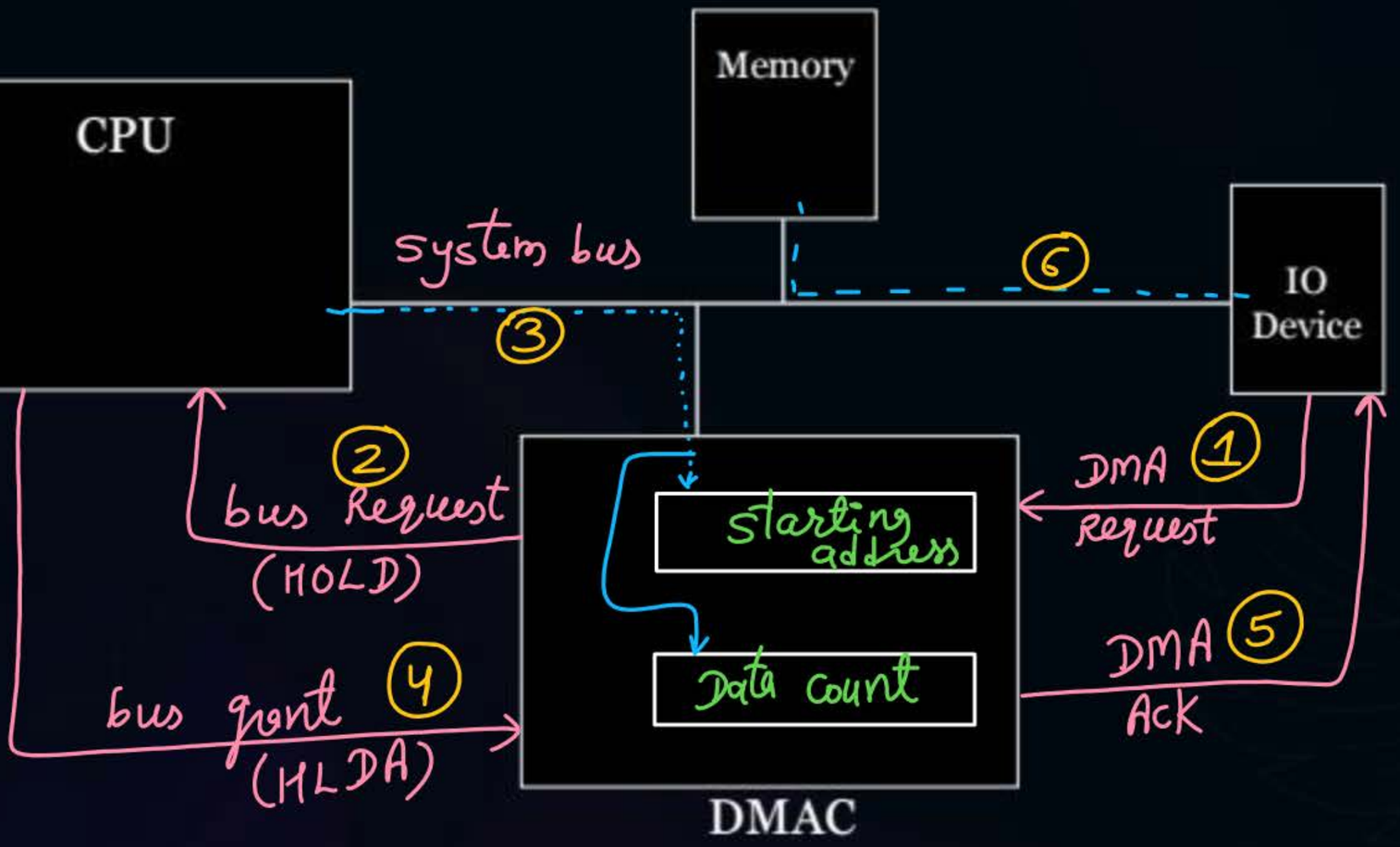


Topic : DMA (Direct Memory Access)

- Enables data transfer between I/O and memory without CPU intervention
- Need a hardware: DMAC (DMA Controller)



Topic : DMA


Memory
CPU
System bus
6
IO Device
3
2
bus Request
(HOLD)
DMA
1
Request
Starting address
Data count
DMA
5
Ack
bus grant
4
(HLDA)
DMAC



Topic : DMA

1. Starting Address :- Address of mem. starting from where data transfer to be done.

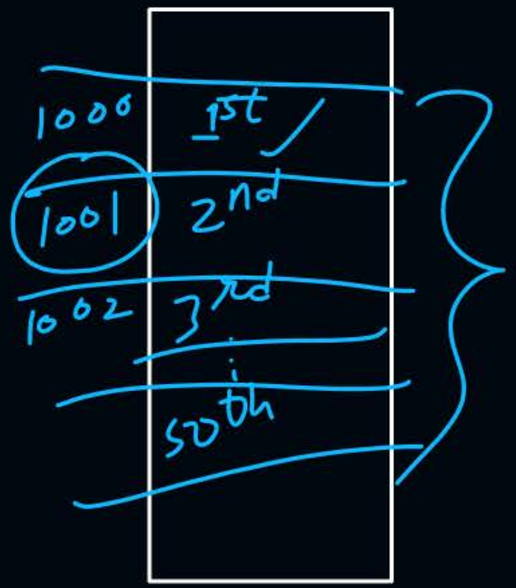
2. Data Count :- no. of bytes/words to be transferred.

for byte addressable memory

for word addressable memory

	Initially	After 1B	After 2B	After 3B		
starting add.	1000	1001	1002	1003	- - - - - -	✕
Data Count	50	49	48	47	- - - - - -	0

data transfer stops



if data count $(1111)_2 = (15)_{10}$

⇓

no. of bytes to be transferred = 15 bytes

Ques) If data count in DMAc has 8-bits. max. no. of bytes DMAC can transfer at a time.

Soln

max binary value with 8 bits $\Rightarrow (11111111)_2$

$\Downarrow$

$(255)_{10}$

Ans = 255



2 mins Summary

Topic Programmed IO

Topic Interrupt IO

Topic DMA



Happy Learning

THANK - YOU